CarbonX

User Manual

Version 0.2

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Contents

Chapter 1

Introduction

Carbon nanotubes (CNTs) exhibit exceptional electrical, optical, and mechanical properties, making them critical candidates for applications in energy storage, sensing, and advanced composites. While chemical vapor deposition (CVD) offers advantages in scalability and cost-effectiveness, controlling CNT morphology in floating-catalyst CVD (FCCVD) reactors remains challenging due to the complex interplay between chemical kinetics, catalyst nanoparticle evolution, and carbon deposition mechanisms. `CarbonX` is an **object-oriented Python package** designed to simulate FCCVD reactors and predict the gas-phase synthesis of single- and multi-walled CNTs alongside metallic (Fe, Ni, and Co) nanoparticles. The framework's modular architecture integrates four fully coupled submodules: a **chemical kinetics submodule** for gas-phase reactions; a **surface kinetics submodule** accounting for catalyst activation, deactivation, and hydrogenation; a **particle dynamics submodule** for tracking nanoparticle evolution through inception, surface growth, coagulation, and sintering; and a **CNT dynamics submodule** for predicting nanotube length, diameter, and wall number. A defining feature of `CarbonX` is its **extensible design**, which allows users to seamlessly integrate **new modules**—such as custom **surface kinetics** or **gas kinetics** mechanisms—into the current code without modifying the core simulation engine. This flexibility enables the adaptation of substrate-based models to the dynamic FCCVD environment. Furthermore, an integrated **machine learning submodule** classifies parametric maps to identify the specific process conditions (e.g., temperature, pressure, and catalyst concentration) required for targeted CNT formation. All submodules have been comprehensively benchmarked against experimental measurements of nanoparticle size distributions and CNT growth kinetics across multiple reactor configurations, providing a validated foundation for predictive modeling.

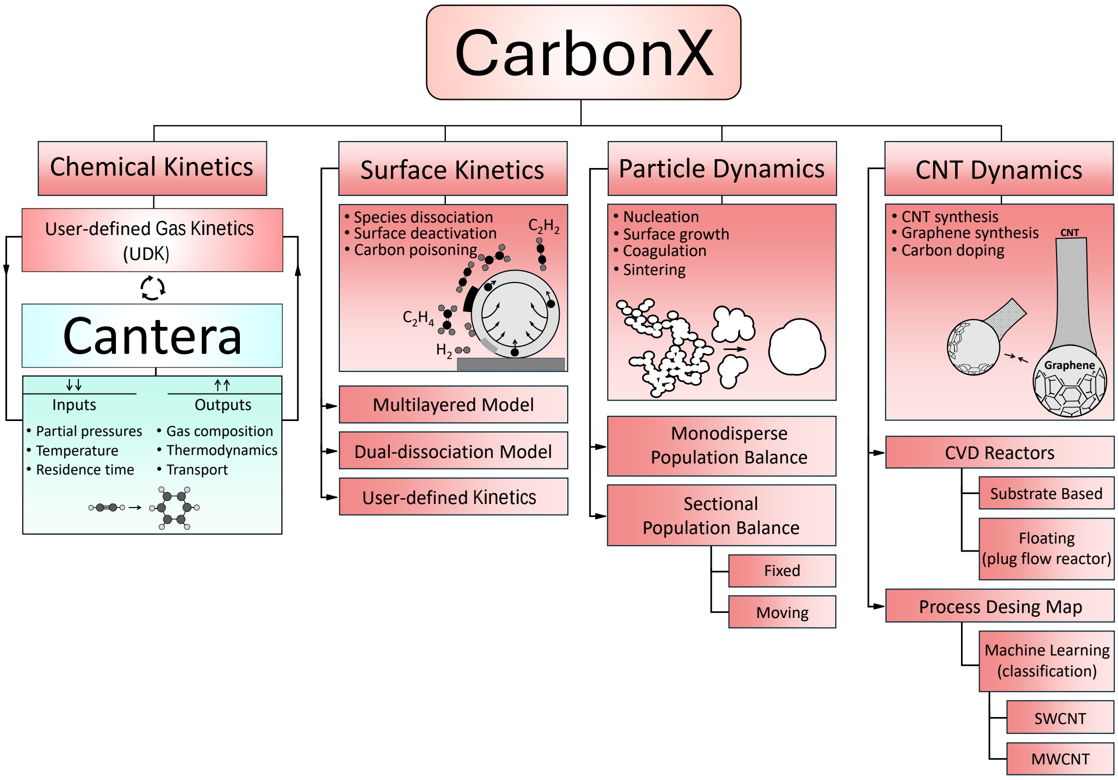

Figure 1.1: Architecture of the `CarbonX` framework for modeling gas-phase synthesis of CNTs in CVD reactors. The package integrates four coupled submodules: chemical kinetics to predict the pyrolysis of the feedstock gas, surface kinetics to predict catalyst-feedstock interactions, particle dynamics to predict the evolution of the morphology of the catalysts, and CNT dynamics to predict CNT inception and characteristics, along with the parametric maps generation.

1.1 Why Python?

`CarbonX` is an object-oriented Python package designed to simulate FCCVD reactors by integrating distinct, fully coupled submodules that handle chemical kinetics, surface kinetics, particle dynamics, and carbon nanotube (CNT) evolution. The choice of Python allows the framework to seamlessly incorporate powerful specialized packages like Cantera for detailed gas-phase analysis and machine learning tools for process optimization, while its extensible design enables users to easily add their own modules, such as custom surface or gas kinetics, without altering the core simulation engine. This modular approach, combined with Python's capacity for continuous integration, ensures the code remains stable and reproducible as researchers adapt the software to model complex interactions between catalyst nanoparticles and carbon deposition across various reactor configurations.

1.2 What can be simulated?

`CarbonX` requires a description of the flow reactor to be simulated (e.g., reactor diameter, length, and temperature history), along with the initial gas-phase chemistry (e.g., carrier gas composition, flow rate, total pressure, and catalyst number concentration). Figure 1.2 outlines the basic workflow.

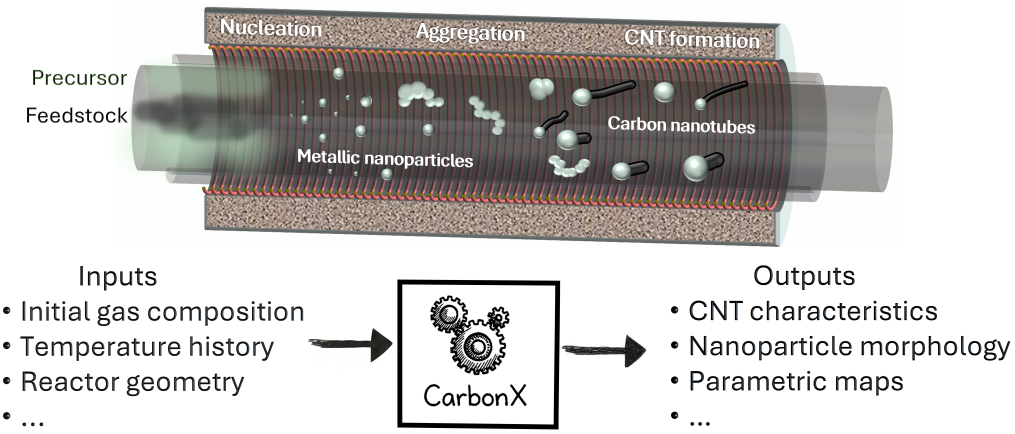

Figure 1.2: Schematic of the CarbonX framework showing key processes (nucleation, aggregation, CNT formation) along with main inputs and outputs.

Chapter 2

What is currently included?

CarbonX is designed as a process design simulation tool and does not aim to model all CNT/graphene phenomena in every reactor configuration produced in every specific reactor. As such, users are responsible for evaluating whether CarbonX is the appropriate tool for their specific application. The *current* version of CarbonX includes a solver to model 1D plug-flow reactors for the production of metallic nanoparticles and carbon nanotubes. The *current* version includes:

- **Modules to simulate the following phenomena:**
  - Agglomeration, surface growth, and sintering of metallic (e.g., Ni, Fe, and Co) nanoparticles.
  - Inception of Fe from iron pentacarbonyl ($Fe(CO)_5$).
  - Pyrolysis of hydrocarbon feedstock gases (e.g., C_2H_2, C_2H_4, CH_4, etc.) using:
    - Available chemical kinetics models (e.g., FFCM-2 [28], Caltech [2], ABF [1]).
    - User-defined detailed chemical kinetics mechanisms.
  - Surface kinetics modeling for:
    - C_2H_2 surface kinetics on Fe nanoparticle surfaces, accounting for adsorption, dissociation, and desorption of C_2H_2, dehydrogenation, as well as nanoparticle carburization, saturation, and deactivation.

    * Custom hydrocarbons on nanoparticle surfaces using user-defined surface kinetics mechanisms.
  - Formation of graphene layers on the surface of nanoparticles.
  - Formation of single-walled (SWCNTs) and multi-walled (MWCNTs) carbon nanotubes on the surface of nanoparticles.

- **Post-processing module to obtain:**
  - Elemental mass balance profiles.
  - Size distribution profiles for nanoparticles and carbon nanotubes.
  - Non-dimensionalized self-preserving size distribution profiles.
  - 2D parametric maps of CNT characteristics, such as diameter, length, and number of walls.
  - Other process parameters (e.g., density, viscosity, species concentration, and nanoparticle carbon impurity), all provided as an output .csv file.

Chapter 3

General Equations

3.1 Chemical kinetics

The chemical kinetics submodule in the `CarbonX` package uses Cantera [6], an open-source suite for reacting flows and thermodynamic calculations. Cantera numerically solves the coupled set of ordinary differential equations (ODEs) governing the time- or space-dependent evolution of species mass fractions in chemically reacting systems. For a homogeneous gas mixture, the mass conservation equation can be written as:

$$\frac{dY_{\mathrm{i}}}{dt} = \frac{\dot{\omega}_{\mathrm{i}} m_{\mathrm{i}}}{\rho_{\mathrm{g}}} \tag{3.1}$$

in which Y_{i} is the mass fraction, $\dot{\omega}_{\mathrm{i}}$ is the net molar production rate per unit volume of gas, ρ_{g} is the density of the carrier gas, and m_{i} is the molecular mass of gas species i.

The term $\dot{\omega}_{\mathrm{i}}$ is calculated based on the elementary reaction mechanism provided as a YAML input file. Each reaction r contributes to the net production rate of gas species i as:

$$\dot{\omega}_{\mathrm{i}} = \sum_{r} \nu_{\mathrm{i,r}} (k_{\mathrm{frw,r}} \prod_{j} [C_{\mathrm{j}}]^{\nu'_{\mathrm{j,r}}} - k_{\mathrm{rev,r}} \prod_{j} [C_{\mathrm{j}}]^{\nu''_{\mathrm{j,r}}}) \tag{3.2}$$

in which $k_{\mathrm{frw,r}}$ and $k_{\mathrm{rev,r}}$ are forward and reverse rate coefficients, $\nu'_{\mathrm{j,r}}$ and $\nu''_{\mathrm{j,r}}$ are the stoichiometric coefficients of reactants and products j, and $\nu_{\mathrm{i,r}}$ is the stoichiometric coefficient of species i, all in the reaction r. While $k_{\mathrm{frw,r}}$ is directly determined from the Arrhenius expression using rate constant parameters from the kinetic mechanism input file, $k_{\mathrm{rev,r}}$ is calculated using $k_{\mathrm{frw,r}}$ and equilibrium constant, $K_{\mathrm{eq,r}}$, which is calculated using the thermodynamic properties given as NASA polynomial coefficients [16], defined for each species in the input mechanism file.

3.2 Particle dynamics: population balance model

The `CarbonX` package employs particle dynamics models (both monodisperse and sectional population balance) [26] to describe the evolution of the morphology of catalyst NPs during inception, surface growth, agglomeration, and sintering in FCCVD reactors, knowing their characteristics such as their primary particle d_{p} and mobility (d_{m}) diameters, and number of primary particle per agglomerate (n_{p}), which are determined by tracking agglomerate number density (N), volume concentration (V), and area concentration (A). The 1D (volume-based) sectional population balance model (SPBM) is mathematically described by the general dynamic equation as [11]:

$$\begin{aligned}\frac{\partial n(v,t)}{\partial t} + \frac{\partial\Big(G(v)\,n(v,t)\Big)}{\partial v} &= \frac{1}{2}\int_0^\infty n(v-v',t)\,n(v',t)\,\beta(v-v',v')\,dv' \\ &\quad - n(v,t)\int_0^\infty n(v',t)\,\beta(v,v')\,dv' + S(v)\end{aligned} \tag{3.3}$$

where $n(v,t)$ represents the **density function** of particles with volume between v and $v+dv$ at time t, $G(v)$ is the size-dependent growth rate, $\beta(v,v')$ denotes the Brownian collision frequency between particles of sizes v and v', and $S(v)$ is the inception rate of particles of size v. Eq. 3.3 can be discretized by dividing the particle-size range into sections (bins), where section i is represented by a representative volume x_i and by the particle number and area concentrations N_i and A_i, respectively.

3.2.1 Agglomeration

During NP agglomeration, the rate of change of N in the section i is calculated as:

$$\begin{aligned}\left.\frac{dN_{\mathrm{i}}}{dt}\right|_{\mathrm{agg}} &= \sum_{x_{\mathrm{i-1}}\le x_{\mathrm{j}}+x_{\mathrm{k}}\le x_{\mathrm{i}}}^{i\ge j\ge k}\left(1-\frac{1}{2}\delta_{\mathrm{j,k}}\right)\beta_{\mathrm{j,k}}N_{\mathrm{j}}N_{\mathrm{k}}\left(\frac{x_{\mathrm{i\text{-}1}}-x_{\mathrm{j}}-x_{\mathrm{k}}}{x_{\mathrm{i\text{-}1}}-x_{\mathrm{i}}}\right)\rho_{\mathrm{g}} \\ &+ \sum_{x_{\mathrm{i}}\le x_{\mathrm{j}}+x_{\mathrm{k}}\le x_{\mathrm{i+1}}}^{i\ge j\ge k}\left(1-\frac{1}{2}\delta_{\mathrm{j,k}}\right)\beta_{\mathrm{j,k}}N_{\mathrm{j}}N_{\mathrm{k}}\left(\frac{x_{\mathrm{i+1}}-x_{\mathrm{j}}-x_{\mathrm{k}}}{x_{\mathrm{i+1}}-x_{\mathrm{i}}}\right)\rho_{\mathrm{g}} - N_{\mathrm{i}}\sum_{k=1}^{B}\beta_{\mathrm{i,k}}N_{\mathrm{k}}\rho_{\mathrm{g}}\end{aligned} \tag{3.4}$$

where B is the total number of sections, $\delta_{\mathrm{j,k}}$ is the Dirac delta function ($\delta_{\mathrm{j,k}} = 1$ if $j = k$, otherwise $\delta_{\mathrm{j,k}} = 0$), and x_{i} is the volume of the pivot in the i^{th} section with boundaries v_i and $v_{\mathrm{i+1}}$ (Figure 3.1). $\beta_{\mathrm{j,k}}$ is the Brownian collision frequency function between particles in the pivots j^{th} and k^{th}, valid across all regimes, from the ballistic (free molecular) to the continuum (diffusive) regimes, and can be expressed as [4]:

$$\beta_{\mathrm{j,k}} = \beta_{\mathrm{co}} \cdot \left[\frac{d_{\mathrm{c,j}} + d_{\mathrm{c,k}}}{d_{\mathrm{c,j}} + d_{\mathrm{c,k}} + 2\bar{\delta}_{\mathrm{r}}} + \frac{8\left(D_{\mathrm{j}} + D_{\mathrm{k}}\right)}{\bar{c}_r\left(d_{\mathrm{c,j}} + d_{\mathrm{c,k}}\right)} \right]^{-1} \tag{3.5}$$

where β_{co} is the collision frequency in the continuum regime with:

$$\beta_{\mathrm{co}} = 4\pi \left(\frac{d_{\mathrm{c,j}}}{2} + \frac{d_{\mathrm{c,k}}}{2} \right) \left(D_{\mathrm{j}} + D_{\mathrm{k}}\right) \tag{3.6}$$

$\bar{\delta}_{\mathrm{r}}$, $\bar{c}_{\mathrm{r}}$, and D are the transition correction factor, Brownian average velocity of particle, and diffusion coefficient, respectively.

The evolution of the particle volume concentration in the section i, V_{i}, is defined as [24]:

$$\left.\frac{dV_{\mathrm{i}}}{dt}\right|_{\mathrm{agg}} = \sum_{x_{\mathrm{i\text{-}1}} \leq x_{\mathrm{j}} + x_{\mathrm{k}} \leq x_{\mathrm{i+1}}}^{i \geq j \geq k} \left(1 - \frac{1}{2}\delta_{\mathrm{j,\,k}}\right) \beta_{\mathrm{j,\,k}} \left(N_{\mathrm{j}} V_{\mathrm{k}} + N_{\mathrm{k}} V_{\mathrm{j}}\right) \eta' \rho_{\mathrm{g}} - V_{\mathrm{i}} \sum_{k=1}^{B} \beta_{\mathrm{i,\,k}} N_{\mathrm{k}} \rho_{\mathrm{g}} \tag{3.7}$$

where η' is the modified volume (interpolation) fraction:

$$\eta' = \begin{cases} \dfrac{\frac{x_{\mathrm{i+1}}}{x_{\mathrm{j}}+x_{\mathrm{k}}} - 1}{\frac{x_{\mathrm{i+1}}}{x_{\mathrm{i}}} - 1}, & x_{\mathrm{i}} \leq x_{\mathrm{j}} + x_{\mathrm{k}} \leq x_{\mathrm{i+1}} \\[2ex] \dfrac{\frac{x_{\mathrm{i\text{-}1}}}{x_{\mathrm{j}}+x_{\mathrm{k}}} - 1}{\frac{x_{\mathrm{i\text{-}1}}}{x_{\mathrm{i}}} - 1}, & x_{\mathrm{i\text{-}1}} \leq x_{\mathrm{j}} + x_{\mathrm{k}} \leq x_{\mathrm{i}} \end{cases} \tag{3.8}$$

The rate of change of agglomerate area in the section i, A_i, is calculated as [24]:

$$\left.\frac{dA_{\mathrm{i}}}{dt}\right|_{\mathrm{agg}} = \sum_{x_{\mathrm{i\text{-}1}} \leq x_{\mathrm{j}} + x_{\mathrm{k}} \leq x_{\mathrm{i+1}}}^{i \geq j \geq k} \left(1 - \frac{1}{2}\delta_{\mathrm{j,\,k}}\right) \beta_{\mathrm{j,\,k}} \left(N_{\mathrm{j}} A_{\mathrm{k}} + N_{\mathrm{k}} A_{\mathrm{j}}\right) \eta' \rho_{\mathrm{g}} - A_{\mathrm{i}} \sum_{k=1}^{B} \beta_{\mathrm{i,\,k}} N_{\mathrm{k}} \rho_{\mathrm{g}} \tag{3.9}$$

3.2.2 Sintering

During the synthesis of NPs in an FCCVD reactor, fusion occurs to minimize surface free energy associated with curvature-induced excess chemical potential [22] while conserving the total volume concentration (V_{i}). The rate of change of A_{i} due to sintering is described as [25]:

$$\left.\frac{dA_{\mathrm{i}}}{dt}\right|_{\mathrm{sin}} = -\frac{1}{\tau_{\mathrm{s,i}}}\left(A_{\mathrm{i}} - N_{\mathrm{i}}\alpha_{\mathrm{i,s}}\right) \tag{3.10}$$

where $\tau_{\mathrm{s,i}}$ is the characteristic sintering time and $\alpha_{\mathrm{i,s}} = a_{\mathrm{o}}\left(V_{\mathrm{i}}/N_{\mathrm{i}}v_0\right)^{2/3}$ is the surface area of a fully sintered (coalesced) agglomerate in the i^{th} section with a_o and v_o denoting the surface area and volume of a monomer, respectively. The parameter τ_{s} is defined as the time required for a 63% reduction in the excess dimer surface area of two fusing primary particles over that of a sphere of equal mass [10]. The τ_{s} for Ni [22] and Fe [23] NPs is given by:

$$\tau_{\mathrm{s,Ni}} = \begin{cases} 6.9\times 10^{15}\cdot T_{\mathrm{p}}d_{\mathrm{p}}^4 \exp\left(\dfrac{64\ 009}{RT_{\mathrm{p}}}\left(2.558 - \left(\dfrac{3.5\times 10^{-9}}{d_{\mathrm{p}}}\times\dfrac{T_{\mathrm{p}}}{2100}\right)^3\right)\right), & T_{\mathrm{p}} < T_{\mathrm{melt}} \\ \dfrac{\mu d_{\mathrm{p}}}{\gamma}, & T_{\mathrm{p}} \geq T_{\mathrm{melt}} \end{cases} \tag{3.11}$$

$$\tau_{\mathrm{s,Fe}} = A_{\mathrm{s,Fe}}d_{\mathrm{p}}^{\mathrm{m}}\exp\left(\frac{E_{\mathrm{a,Fe}}}{RT_{\mathrm{p}}}\right) \tag{3.12}$$

where $d_{\mathrm{p}} = 6V_{\mathrm{i}}/A_{\mathrm{i}}$ is the primary particle diameter, μ [17] and γ [7] are the dynamic viscosity and surface tension of Ni, respectively, and $T_{\mathrm{melt}} = -1.18\times 10^3\cdot d_{\mathrm{p}}^{-0.9} + 1836$ K is the size-dependent melting temperature of Ni NPs [22]. $A_{\mathrm{s,Fe}}$ and $E_{\mathrm{a,Fe}}$ are the pre-exponential factor and activation energy, respectively. The exponent m describes the dominant mass transfer mechanism (i.e., grain-boundary diffusion, surface diffusion, and viscous flow) during the sintering process [13]. The particle temperature, T_{p}, is distinguished from that of the carrier gas temperature, T, by introducing [12]:

$$c_{\mathrm{v}}n_{\mathrm{p}}\frac{dT_{\mathrm{p}}}{dt} = \frac{\gamma}{\tau_{\mathrm{s}}}\left(\alpha - \alpha_{\mathrm{s}}\right) - Zc_{\mathrm{g}}\left(T_{\mathrm{p}} - T\right) - \varepsilon_{\mathrm{p}}\sigma_{\mathrm{SB}}\alpha_{\mathrm{s}}\left(T_{\mathrm{p}}^4 - T^4\right) \tag{3.13}$$

where c_{v} and c_{g} are the specific heats for the particle and carrier gas, respectively, ε_{p} ($\sim$0.5) is the emissivity of the particle [27], σ_{SB} is the Stefan-Boltzmann constant, and $Z = \alpha_{\mathrm{s}}\mathrm{P}/(2\pi m_{\mathrm{g}}k_{\mathrm{B}}T)^{0.5}$ is the collision rate between particles and carrier gas molecules with P representing the reactor pressure and m_{g} the carrier gas molecular mass. The first term on the right-hand side (RHS) in Eq. 3.13 represents an increase in the particle temperature due to the energy released during coalescence, during which the particle SSA decreases. The second and third RHS terms describe particle cooling by convective/conductive heat transfer and radiative losses, respectively.

3.2.3 Inception and surface growth

The rates of change of N, V, and A due to the inception (i.e., formation of metal atoms from precursor decomposition) of primary particles and their surface growth (i.e., heterogeneous atom deposition on primary particle surfaces) are defined as:

$$\left.\frac{dN_\mathrm{i}}{dt}\right|_\mathrm{incep} = k_\mathrm{g} C n_\mathrm{i} \quad \text{with} \quad n_\mathrm{i} = \begin{cases} 1 & v_\mathrm{m} \in [v_\mathrm{i}, v_\mathrm{i+1}] \\ 0 & v_\mathrm{m} \notin [v_\mathrm{i}, v_\mathrm{i+1}] \end{cases} \tag{3.14}$$

$$\left.\frac{dA_\mathrm{i}}{dt}\right|_\mathrm{incep+surf} = k_\mathrm{g} C \eta_\mathrm{i} \alpha_\mathrm{m} + 4\pi N_\mathrm{i} n_\mathrm{p,i} k_\mathrm{s} d_\mathrm{p,i} C v_\mathrm{m} \rho_\mathrm{g} \tag{3.15}$$

$$\frac{dx_\mathrm{i}}{dt}\Big|_\mathrm{incep+surf} = k_\mathrm{s}\, C\, n_\mathrm{p,i}\, \pi d_\mathrm{p,i}^2\, v_\mathrm{m}\, \rho_\mathrm{g} \; + \; \frac{1}{N_\mathrm{i}}\, (v_\mathrm{m} - x_\mathrm{i})\, (k_\mathrm{g} C)\, \eta_\mathrm{i} \tag{3.16}$$

$$\left.\frac{dV_\mathrm{i}}{dt}\right|_\mathrm{incep+surf} = \left.\frac{dN_\mathrm{i} x_\mathrm{i}}{dt}\right|_\mathrm{incep+surf} \tag{3.17}$$

where C is the precursor concentration with its homogeneous and heterogeneous decomposition rate constants of k_g and k_s, respectively. Additionally, v_m is the volume and α_m is the area of a single metallic atom.

3.3 Surface kinetics

The *current* version includes two surface kinetics models (i.e., multilayered and dual−dissociation) for thermal decomposition of C_2H_2 on the surface of Fe nanoparticles.

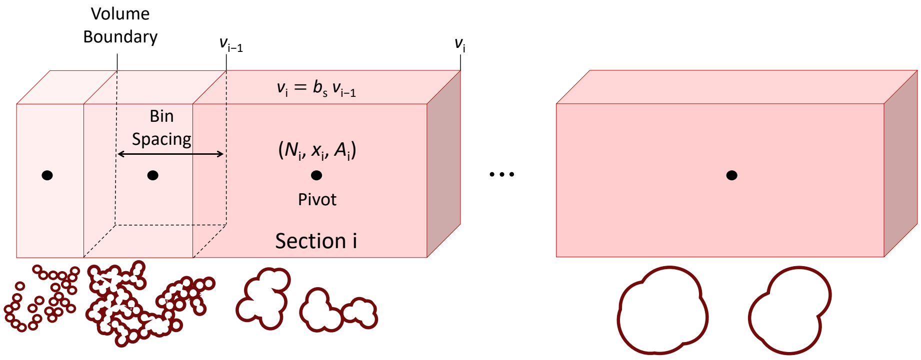

Figure 3.1: Schematic of the sectional method for particle population balance modeling. Volume domain is discretized into sections with the spacing factor b_s, tracking number concentration (N_i), aggregate volume (x_i), and surface area (A_i) per section. Particle morphology evolves from fractal aggregates to spherical structures through sintering.

3.3.1 Multilayered model

This model was originally proposed by Puretzky et al. [21] to describe kinetic data measured in a substrate-based CVD reactor using acetylene (C_2H_2) as the carbon feedstock and iron-molybdenum NPs as the catalysts. The model accounts for activation energies and rate constants associated with surface carbon formation and diffusion into the NPs. The number of carbon atoms incorporated into a CNT (N_t) is determined by tracking several carbon-related populations on the surface of the NPs: (i) carbon atoms on the surface of the NPs (N_C), formed due to catalytic decomposition of C_2H_2; (ii) carbon atoms converted into the carbonaceous (platelet) layer on the NP surface (N_{L1}), (iii) inactive catalyst atoms on the NP surface (N_{L2}), and (iv) carbon atoms diffusing into the bulk region of NPs (N_B). The schematic in Figure 3.2 illustrates the model, accounting for the adsorption of C_2H_2, carbon surface and bulk diffusion, formation of carbon platelets, and hydrogasification process ($C_s + 2H_2 \rightarrow CH_4$), along with the inception of graphene layers and CNT on the surface of NPs. The C_2H_2 flux, F_{b1}, and its main pyrolysis products (e.g., C_4H_4 [15]), F_{b2}, toward the surface of NPs within section i are given by:

$$F_{b1} = \frac{1}{4} A_p q \left(\frac{k_B T}{2\pi m} \right)^{1/2} \tag{3.18}$$

$$F_{b2} = \frac{1}{4} A_p q_p \left(\frac{k_B T}{2\pi M} \right)^{1/2} \tag{3.19}$$

where $A_p = A_i/N_i$ is the surface area of one agglomerate in section i, q, q_p are the number concentrations of the C_2H_2 and its main pyrolysis product (e.g., C_4H_4) in the carrier gas mixture, and m and M are the masses of C_2H_2 molecule and its major pyrolysis product, respectively. The corresponding carbon fluxes due to F_{b1} and F_{b2} are:

$$F_{c1} = F_{b1} p_1 \exp\left(-\frac{E_{a1}}{RT}\right) \tag{3.20}$$

$$F_{c2} = F_{b2} p_2 \exp\left(-\frac{E_{a2}}{RT}\right) \tag{3.21}$$

where $p_1 = p_2 = 1$ are the pre-exponential factors and $E_{a1} = E_{a2} = 57\ 891$ J/mol are the activation energies for catalytic decomposition of the C_2H_2 and its major products, respectively [21]. The processes discussed above can be described by the following set of differential equations as:

$$\frac{dN_C}{dt} = F_{c1}\left(1 - \frac{N_{L1} + N_{L2}}{\alpha A_p n_m}\right) - (k_{sb} + k_{cl} + k_e) N_C \tag{3.22}$$

$$\frac{dN_{L1}}{dt} = F_{c2}\left(1 - \frac{N_{L1} + N_{L2}}{\alpha A_p n_m}\right) + k_{cl} N_C - (k_{d1} + k_e) N_{L1} \tag{3.23}$$

$$\frac{dN_{L2}}{dt} = k_r\left(1 - \frac{N_{L1} + N_{L2}}{\alpha A_p n_m}\right) - k_{d2} N_{L2} \tag{3.24}$$

$$\frac{dN_B}{dt} = k_{s\ b} N_C - k_t N_B + k_{d1} N_{L1} \tag{3.25}$$

$$\frac{dN_t}{dt} = k_t N_B \tag{3.26}$$

where $n_m = 10^{19}$ #/m^2 and α [21] are the surface density and number of carbon monolayers, respectively, k_{sb} the rate constant for surface–to-bulk penetration of carbon atoms, k_{cl} the rate for platelet formation, k_e the rate of carbon atom removal due to methane (CH_4) formation through the etching process [5], and k_t is the rate for carbon diffusion to the

CNT/graphene growth sites region with [21]:

$$k_{\mathrm{sb}} = 2.72 \times 10^{14} \exp\left(-\frac{212\ 267}{RT}\right) \tag{3.27}$$

$$k_{\mathrm{cl}} \approx 3 \times 10^{-3} \tag{3.28}$$

$$k_{\mathrm{e}} = \frac{(1.15 \times 10^{-13}) \exp\left(-\frac{72\ 100}{RT}\right) P_{\mathrm{H_2}}^2}{1 + (3.50 \times 10^{-12}) \exp\left(-\frac{118\ 100}{RT}\right) P_{\mathrm{H_2}}} \tag{3.29}$$

$$k_{\mathrm{t}} = \left(\frac{2 \times 10^{-4}}{d_{\mathrm{p}}^2}\right) \exp\left(-\frac{144\ 728}{RT}\right) \tag{3.30}$$

Here, k_{d1} and k_{d2} are the rate constants corresponding to the dissolution of the platelets and the reactivation of the deactivated layer (caused by the presence of water), respectively, and both are assumed to be negligible (i.e., $k_{\mathrm{d1}} = k_{\mathrm{d2}} = 0$) [21]. The rate of catalyst deactivation due to H_2 diffusion, k_{r}, can also be expressed as:

$$k_{\mathrm{r}} = 1.2 \times 10^{8} F_{\mathrm{H_2}} \exp\left(-\frac{231\ 564}{RT}\right) \tag{3.31}$$

where the H_2 flux into the NP ($F_{\mathrm{H_2}}$) can be estimated from its partial pressure ($P_{\mathrm{H_2}}$) as:

$$F_{\mathrm{H_2}} = 2\alpha_{\mathrm{H_2}} D_{\mathrm{H_2}} \frac{N_{\mathrm{Avo}} P_{\mathrm{H_2}}}{RT} \frac{A}{d_{\mathrm{p}}} \tag{3.32}$$

with a sticking probability of $\alpha_{\mathrm{H_2}} = 7 \times 10^{-4}$ and an H_2 diffusion coefficient of $D_{\mathrm{H_2}} = 5 \times 10^{-8} \exp\left(-\frac{10\ 958}{RT}\right)$. The number of walls, N_{w}, can be determined from the thickness of the disordered layer (Figure 3.2a) expressed as the number of carbon monolayers:

$$N_{\mathrm{w}} = \begin{cases} N_{\mathrm{C}}/\tilde{\alpha} A_{\mathrm{p}} n_{\mathrm{m}} & \text{if } N_{\mathrm{w}} > 1 \\ 1 & \text{if } N_{\mathrm{w}} \leq 1 \end{cases} \tag{3.33}$$

where $\tilde{\alpha}$ is an adjustable parameter tuned to the experimental data, with $\tilde{\alpha} = 0.3$ [21].

3.3.2 Dual-dissociation model

This model was developed by Ma *et al.* [15] to predict the growth rate of MWCNTs in a substrate-based CVD reactor from the decomposition of C_2H_2 and C_4H_4 (the major decomposition product). Three key steps govern the synthesis process under continuous feedstock supply: (i) adsorption and dissociation of C_2H_2 and C_4H_4 on the Fe NP surface, i.e., $C_2H_2 \rightarrow C_{2,s} + H_2$ and $C_4H_4 \rightarrow C_{4,s} + 2H_2$, where the subscript "s" denotes species on the NP surface; (ii) carbon–metal solid solution or carbide formation; and (iii) carbon precipitation from supersaturated NPs (i.e., $C_{2,s} \rightarrow 2C_{cnt}$ and $C_{4,s} \rightarrow 4C_{cnt}$). Therefore, the number of carbon atoms precipitated into the CNT (N_t) is obtained by tracking the number of carbon atoms dissolved in the NP (N_B) as:

$$\frac{dN_t}{dt} = \frac{dN_B}{dt} \tag{3.34}$$

$$\frac{dN_B}{dt} = A_p \gamma_p \rho_{gas} \sqrt{\frac{k_B T}{2\pi}} \frac{[O_s]}{[O_{s,o}]} \left(\frac{[C_{C_2H_2}]}{\sqrt{m}} + \frac{[C_{C_4H_4}]}{\sqrt{M}} \right) \tag{3.35}$$

where γ_p is the sticking probability of reactant species adsorbing onto the surface of NPs, $[C_nH_m]$, $[O_{s,o}]$, and $[O_s]$ are the concentrations of C_nH_m, as well as the initial and current active site concentrations on the NP surface, respectively:

$$\gamma_p = \begin{cases} 6.38 \times 10^{-3} \exp(1.11 \times 10^{-2}(T - 823.81)), & T \leq 973 \\ 3.46 \times 10^{-2} \exp(-3.27 \times 10^{-2}(T - 973)), & T > 973 \end{cases} \tag{3.36}$$

Although one of the advantages of the dual-dissociation model is its ability to track the concentration of adsorption sites, it does not predict the number of walls due to the lack of information about the thickness of the disordered layer. Instead, as an explicit assumption, the model assumes the inner diameter of the CNT is half of its outer diameter, i.e., $d_{cnt,inner} = 0.5\, d_{cnt,outer}$. In this package, the outer diameter of the CNT produced in an FCCVD reactor is calculated from the primary particle diameter of the NPs, as statistical measurements by Nasibulin et al. [19] showed that the ratio between the catalyst and the CNT diameters remains close to 1.6 (i.e., $d_{cnt} \approx d_p/1.6$), regardless of process conditions.

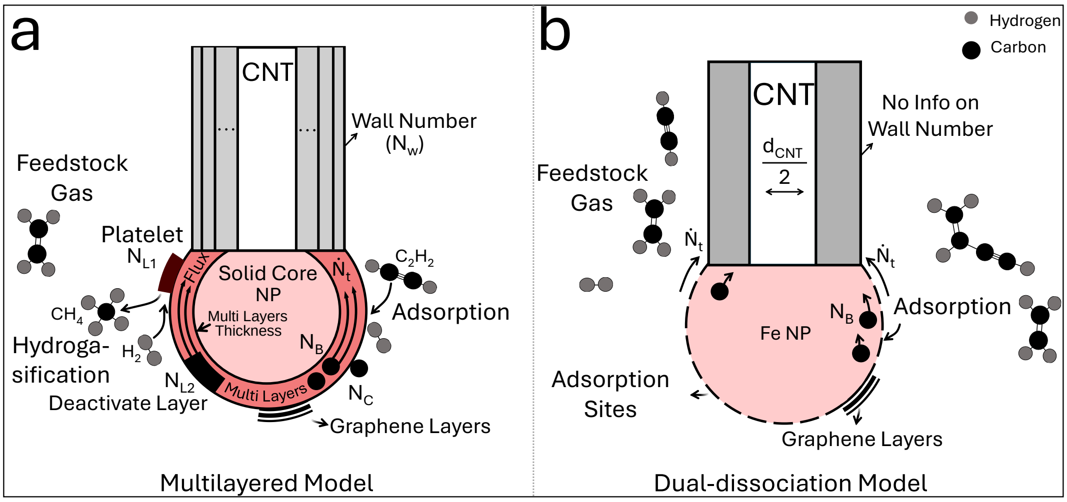

Figure 3.2: Schematic of two surface kinetics models implemented in the `CarbonX` package: (a) the multilayered model, which predicts the carbon precipitation rate at the CNT growth site ($\dot{N}_\mathrm{t}$) by tracking multiple carbon-related populations, including carbon atoms on the NP surface (N_C), in the NP bulk region (N_B), and in the platelet layers (N_L1); (b) the dual-dissociation model, which predicts $\dot{N}_\mathrm{t}$ by tracking the bulk carbon population (N_B) of NPs.

3.3.3 User-defined kinetics (UDK)

To further enhance the flexibility of the package, a user-defined kinetics (UDK) submodule is included, enabling the integration of custom surface kinetics models. The user-defined reaction pathways can be specified in a YAML input file, following the standard format used for surface mechanisms, including site adsorption/desorption, hydrogen abstraction, and carbon diffusion (e.g., Table 1 in [14]). The submodule enables full coupling between user-specified surface reactions and the background gas chemistry solved by the chemical kinetics submodule.

3.4 CNT dynamics

Before the onset of CNT in the reactor, carbon is assumed to dissolve into the nanoparticles until it reaches saturation, beyond which no additional carbon can dissolve [19]. The expressions for size-dependent carbon solubility of Ni and Fe NPs were proposed by Moisala et al. [18] as:

$$\frac{\mathrm{N_{sat}}}{\mathrm{N_{Ni}}} = \left(0.01775 + 5.219 \times 10^{-5}\ T\right) \exp\left(\frac{4\nu_\mathrm{Ni}\gamma_\mathrm{Ni}}{k_\mathrm{B} T \mathrm{d_p}}\right) \tag{3.37}$$

$$\frac{\mathrm{N_{sat}}}{\mathrm{N_{Fe}}} = \left(0.062305 + 1.176 \times 10^{-4}\ T\right) \exp\left(\frac{4\nu_{\mathrm{Fe}}\gamma_{\mathrm{Fe}}}{k_{\mathrm{B}}T\mathrm{d_p}}\right) \tag{3.38}$$

where N_{Ni} and N_{Fe} are the number of metal atoms (monomers) in a primary particle, $\gamma_{\mathrm{Ni}} = 2$ [7] and $\gamma_{\mathrm{Fe}} = 1.72$ [18] are surface tensions, and ν_{Ni} and ν_{Fe} are the atomic volumes for Ni and Fe, respectively.

From Eqs. 3.26 and 3.34, under carbon-saturated conditions, the rates of carbon deposition into the CNT and graphene layers can be defined as [3]:

$$\frac{dL_{\mathrm{cnt}}}{dt} = f_{\mathrm{cnt}}\left(\frac{m_{\mathrm{c}}}{\pi\rho_{\mathrm{c}}\delta_{\mathrm{gp}}d_{\mathrm{cnt}}}\right)\frac{dN_{\mathrm{t}}}{dt} \tag{3.39}$$

$$\frac{dA_{\mathrm{gp}}}{dt} = (1 - f_{\mathrm{cnt}})\left(\frac{m_{\mathrm{c}}}{\pi\rho_{\mathrm{c}}\delta_{\mathrm{gp}}}\right)\frac{dN_{\mathrm{t}}}{dt} \tag{3.40}$$

where f_{cnt} is the fraction of carbon atoms (N_{t}) that contribute to the formation of CNTs, m_{c} is the mass of one carbon atom, $\rho_{\mathrm{c}} = 1.8$ g/cm^3 and $\delta_{\mathrm{gp}} = 0.34$ nm are the graphite density and graphene layer thickness, respectively [20]. The mobility diameter of agglomerates can be correlated with the average number of primary particles per agglomerate, valid for all Knudsen regimes:

$$d_{\mathrm{m}}^{\mathrm{NP}} = d_{\mathrm{p}}n_{\mathrm{p}}^{0.45} \tag{3.41}$$

where the average number of primary particles in each agglomerate, n_{p}, is [8]:

$$n_{\mathrm{p}} = \frac{6V}{N\pi d_{\mathrm{p}}^3} \tag{3.42}$$

Also, the gyration diameter, $d_{\mathrm{g}}^{\mathrm{NP}}$, is calculated as [8]:

$$d_{\mathrm{g}}^{\mathrm{NP}} = \begin{cases} d_{\mathrm{m}}^{\mathrm{NP}}\left(n_{\mathrm{p}}^{-0.2} + 0.4\right)^{-1}, & n_{\mathrm{p}} > 1.8 \\ d_{\mathrm{m}}^{\mathrm{NP}}\sqrt{\frac{3}{5}}, & n_{\mathrm{p}} \leq 1.8 \end{cases} \tag{3.43}$$

From tandem differential mobility analyzer measurements, Kim and Zachariah [9] proposed an expression for the on-the-fly calculation of CNT mobility diameter (all sizes in nm) as:

$$d_{\mathrm{m}}^{\mathrm{cnt}} = 8.61 \times 10^{-1}\sqrt{\frac{4d_{\mathrm{cnt}}\mathrm{L_{cnt}}}{\pi}} + 7.6 \tag{3.44}$$

3.5 Iteration scheme

Figure 3.3 illustrates the workflow of the `CarbonX` package, which consists of three main components: input (steps 1-3), reactor modeling (steps 4-10), and output handling (step 11). The key components of the user input include chemical kinetics (FFCM-2, Caltech, ABF, etc.), surface kinetics (multilayered, dual-dissociation, and user-defined), and particle dynamics model name specifications, as well as general configuration settings (e.g., total number of simulations along with residence time, temperature-time history, flow rates, reactor pressure, etc.). In steps 4-5, the gas chemistry of the system is solved using the Cantera-UDK submodule, where thermodynamic (e.g., density and pressure) and transport (e.g., viscosity) properties, and species mole fractions are obtained incrementally.

Current NP population and morphology data are passed to the surface kinetics submodule to calculate carbon fluxes in the bulk, multilayered, and CNT site regions. The states of NPs in terms of encapsulation and poisoning are also evaluated. In step 7, the NP morphology is updated within each bin, followed by steps 8-9, where the CNT and graphene characteristics (i.e., d_{cnt}, L_{cnt}, and A_{gp}) are estimated.

Once a simulation cycle is completed (step 10), the process is repeated until all points in the 2D characteristic maps are generated across the selected (range of) process conditions (e.g., $P_{\mathrm{C_2H_2}}$ vs. T, N vs. T, and d_{p} vs. T). Finally, the machine-learning-based classification submodule is activated (step 11) to classify the predicted maps into regions corresponding to the process conditions that yield SWCNTs and MWCNTs in FCCVD reactors.

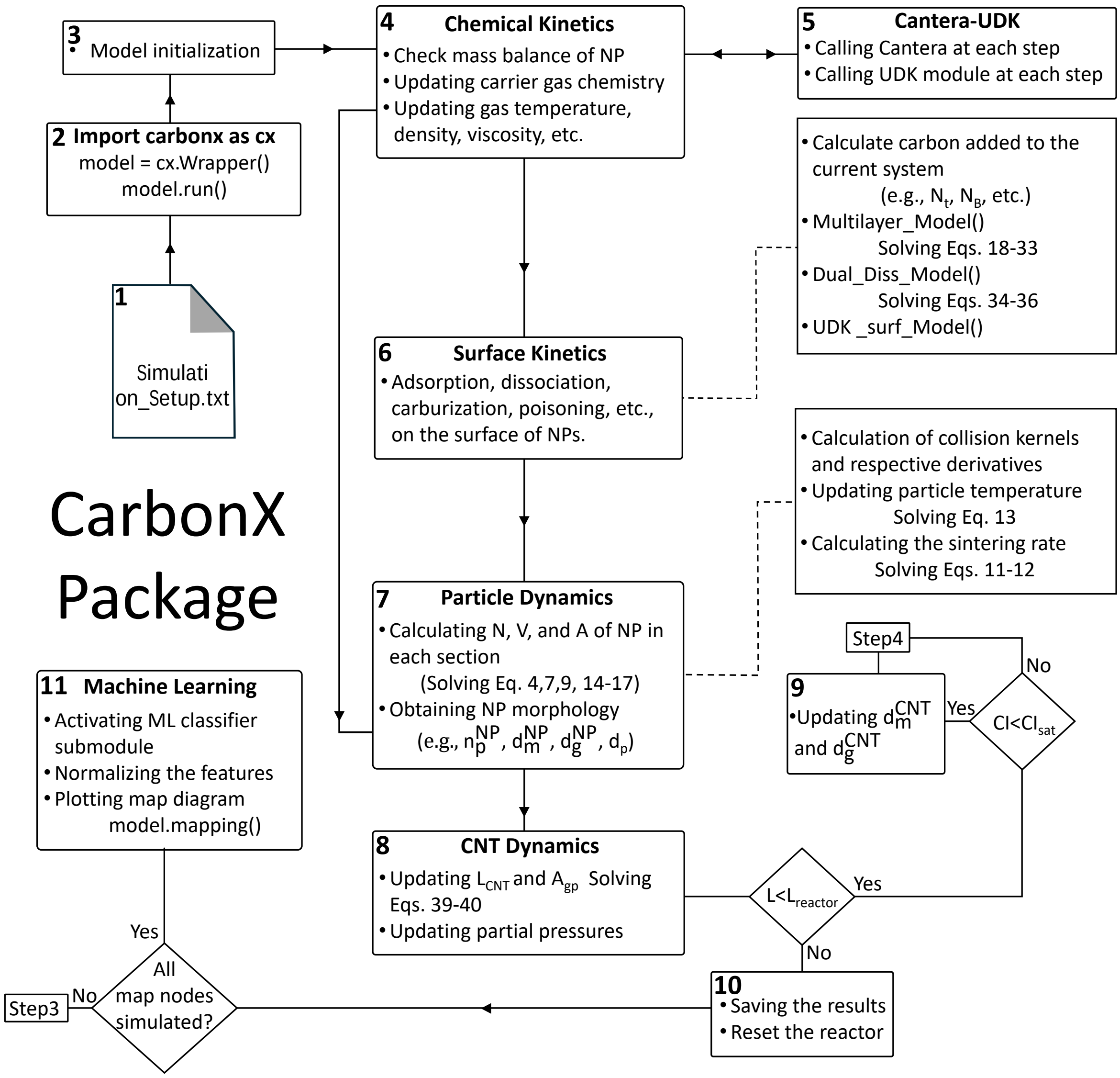

Figure 3.3: The architecture of `CarbonX`. The framework integrates four fully coupled components: (1) a chemical kinetics submodule coupled with UDK to compute thermodynamic and transport properties; (2) a surface kinetics submodule to track carbon concentrations and NP poisoning/encapsulation states; (3) a particle dynamics submodule employing population-balance models to update NP morphology; and (4) a CNT dynamics submodule to estimate nanotube (L_{cnt} and d_{cnt}) and graphene (A_{gp}) characteristics. A machine-learning-based classification submodule subsequently partitions the obtained 2D maps of CNT characteristics (d_{cnt} and L_{cnt}) into SWCNT and MWCNT formation zones.

Chapter 4

Installation, Input, and Parameters Description

4.1 Installation

`CarbonX` 0.2.6 is available on the Python Package Index (PyPI) at https://pypi.org/project/CarbonX. The package can be installed using the `pip` package manager by running the following command in the terminal or command prompt:

```
pip install carbonx
```

Alternatively, users working in an IDE can install it directly through the built-in package manager. All required dependencies are resolved and installed automatically.

4.1.1 System Requirements

Before installation, ensure your system meets the following requirements:

- **Python Version:** Python 3.11 (required)
- **Operating System:** Windows 64-bit (Linux/macOS support planned for future releases)
- **Architecture:** 64-bit system

Important: The current version of `CarbonX` is compiled specifically for Python 3.11.

Using other Python versions (3.10, 3.12, etc.) will result in installation failure or runtime errors.

4.1.2 Verifying Python Version

To verify your Python version, run the following command:

```
python --version
```

The output should show `Python 3.11.x`. If you have a different version, download and install Python 3.11 from https://www.python.org/downloads/.

4.1.3 Installation Steps

Step 1: Create a Virtual Environment (Recommended)

It is strongly recommended to install `CarbonX` in a virtual environment to avoid dependency conflicts:

```
python -m venv carbonx_env
carbonx_env\Scripts\activate
```

Step 2: Install CarbonX

With the virtual environment activated, install `CarbonX`:

```
pip install carbonx
```

This command automatically installs all required dependencies:

- `numpy` ($\geq$ 1.24.0, $<$ 2.0.0)
- `scipy` ($\geq$ 1.10.0, $<$ 1.14.0)
- `matplotlib` ($\geq$ 3.5.0)
- `pandas` ($\geq$ 1.5.0, $<$ 2.1.0)
- `cantera` ($\geq$ 2.6.0) — chemical kinetics library
- `pyyaml` ($\geq$ 6.0)

4.2 Input

The `CarbonX` package requires two primary objects, depending on the simulation mode. For a single reactor simulation, the `GasReactor` class is used, while `MappingWrapper` is used for multiple reactor conditions, which is the basis for parametric map generation (see the next chapter for details). Although several internal class attributes are accessible directly, it is recommended to instantiate the simulation through these two classes. Additional usage examples are provided in the following chapter. A full description of all controlling parameters is discussed in the following sections.

A minimal working example for a single reactor simulation is shown below:

Listing 4.1: Single reactor simulation using `GasReactor` class.

```
from pathlib import Path
from carbonx import GasReactor
from carbonx.modules.simulation_setup_loader import build_kwargs

SETUP_FILE = Path("simulation_setup.txt")

model = GasReactor(
    **build_kwargs(
        SETUP_FILE,
        catalsyt_element="Ni",
        intnum= 37,
        bin_spacing=1.9,
        rtol=1e-12,
        atol= 1e-38,
        length_step = 'flex_loose',
        kernel_type="fuchs",
        wrapper_mapping_temp=None,
        temperature_history="custom",
        total_initial_concentration=1e+11,
        E_a1=0.9,
        __xqtot=2.01e-5,
```

```
        reactor_length=0.6,
        xdtube=0.0254,
        gas_initial_composition={"C2H2": 0.0045, "H2": 0.045, "N2
           ": 1 - 0.0045- 0.045},
        dp_initial_premade=15e-9,
        surface_kinetics_solver_activated=True,
        carb_struct_enabled=True,
        surface_kinetics_type="Multilayerd_Model",
    )
)
_, solutions = model.run()
```

For parametric map generation over multiple reactor conditions, the `MappingWrapper` class is used as follows:

Listing 4.2: Multiple reactor simulation using `MappingWrapper`.

```
from pathlib import Path
from carbonx import MappingWrapper
from carbonx.modules.simulation_setup_loader import build_kwargs

SETUP_FILE = Path("simulation_setup.txt")

model = MappingWrapper (map_requested="P&T",
                          grid_total =1025,
                          P_range_min =0.001 ,
                          P_range_max =0.01 ,
                          P_iso =0.01 ,
                          T_range_min =800 ,
                          T_range_max =1200 ,
                          T_iso =873 ,
                          xdp_range_min =10e-9 ,
                          xdp_range_max =50e-9 ,
                          xdp_const =15e-9 ,
                          L_reactor_range_min =0.1 ,
                          L_reactor_range_max =1 ,
                          L_reactor =0.6 ,
                          xN_range_min =1e17 ,
                          xN_range_max =10e17 ,
                          xN_const =1e11 ,
                          scale_min =10e-8 ,
                          scale_max =7e-6 ,
                          ml_method = " mean " ,
                          ml_lambda_ =0.001 ,
                          ml_iterations =10000 ,
                          ml_alpha =1.5 ,
                          surface_kinetics_type = "
                             multilayered_model " ,# "
                             multilayered_model"
                          ml_post_cond = True)
```

```
_, solutions = model.run()



results = model.run_parametric_study()
model.parametricstudyvisualizer()
```

All parameters can be set either in the `simulation_setup.txt` file or defined directly in the simulation script. When a parameter is defined in both, the package prioritizes the directly defined value and ignores the one specified in the file. The `simulation_setup.txt` file must be located in the same directory as the simulation script. Any line beginning with a hashtag (`#`) is treated as a comment and is ignored by the package. An example of a `simulation_setup.txt` file is shown below:

Listing 4.3: Example content of `simulation_setup.txt`.

```
###CarbonX###
# Version 0.2
# Definition of temperature history
# Notes: Units:
# T_react: reactor wall temperature in K
# L_react: reactor length in m
#------------------------------------------------------------
T_react=[300, 460, 480, 660, 800, 900]
L_react=[0, .1, .22, .4, .65, .9]
catalyst_element="Fe"
surface_kinetics_solver_activated=False
xdtube=0.02
```

4.3 Description of Simulation Parameters

4.3.1 `GasReactor` Class Parameters

Reactor Configuration

- `reactor_length`: Total length of the reactor in m.

- `__xqtot`: Total volumetric flow rate of the reactor in m^3/s.
- `xdtube`: Reactor diameter in m.
- `total_pressure`: Total reactor pressure in Pa. Note: Only isobaric conditions can be modeled in the current version.
- `kinetics_mechanism_type`: Specifies the gas kinetics model. Available built-in models are `"FFCM-2"`, `"ABF"`, `"Caltech"`, `"KAUST"`, and `"GRI-Mech 3.0"`. Alternatively, a user-defined global or elementary reaction mechanism can be provided in the accepted format (see the `CarbonX` paper for details).
- `gas_kinetics_mechanism_UDF`: Set to `True` if an additional user-defined gas kinetics model is to be coupled with one of the available mechanisms. Default: `False`.
- `gas_initial_composition`: Initial molar fractions of gas species in the reactor, defined as a dictionary. For example:

```
gas_initial_composition={"C2H2": 0.0045, "N2": 1 - 0.0045}
```

  If the total molar fractions sum to less than one, the package normalizes the reactor atmosphere by adjusting the carrier gas defined by the `carrier_species` parameter (default: `"N2"`).
- `gamma`: Sticking probability of the feedstock gas (C_2H_2 in the current version), unitless.
- `surface_kinetics_type`: Type of surface kinetics model. Available options are `"Multilayered_Model"` and `"Dual_Diss_Model_Steady"`. A user-defined surface kinetics model can also be integrated using `"Surface_Kinetics_General_UDF"` in the accepted format (see the CarbonX paper for details).
- `surface_kinetics_solver_activated`: Set to `False` (default) if the reactor is designed to produce metallic nanoparticles only, with no CNT or graphene formation. Set to `True` to activate the surface kinetics solver.
- `carb_struct_enabled`: Set to `False` (default) if the reactor is designed to model metallic nanoparticles in the absence of CNTs or any other carbonaceous species.

Particle Configuration

- `catalyst_element`: The catalyst element to simulate. `CarbonX` 0.2 supports iron (`"Fe"`), nickel (`"Ni"`), and cobalt (`"Co"`) nanoparticles. Additional elements will be added in future versions.

- `total_initial_concentration`: Initial number concentration of the injected catalyst nanoparticles in #/m^3.

- `dp_initial_premade`: Initial primary particle diameter of the injected catalyst in m.

- `initial_distribution`: Distribution of injected nanoparticles. Options are `"Monodisperse"` (default) or `"Polydisperse"`. When `"Polydisperse"` is selected, the following parameters must also be defined: `geometric_std`, `distribution_type` (`"lognormal"` or `"normal"`), `low_PP_range`, and `high_PP_range`.

- `intnum`: Number of sections (bins) used in the sectional population balance model. A higher number of bins increases resolution but also increases computational cost.

- `kernel_type`: Collision kernel type for particle collisions. `"fuchs"` applies the Fuchs interpolation [4], which covers the free molecular, transition, and continuum regimes and is the recommended default. `"cr"` and `"fmr"` are used for the continuum and free molecular regimes, respectively.

- `regrid_condition`: Switches between `"Moving"` and `"Fixed"` sectional population balance models.

- `bin_spacing`: Spacing between sections when the `"Fixed"` sectional model is used. Default: `1.6`.

- `elimination_spacing`: Threshold value for section elimination when the `"Moving"` condition is active. Default: `1.1`.

- `fraction_cnt_gp`: Fraction of carbon atoms incorporated into CNTs. Default: `0.9`.

Other Configuration

- `length_step`: Controls the spatial step size along the reactor, accounting for the stiffness of the simulation as particles evolve. Three options are available: `"flex_tight"` (flexible steps from 10^{-7} to 10^{-3} m), `"flex_extra_tight"` (flexible steps from 10^{-8} to 10^{-3} m), and `"flex_loose"` (flexible steps from 10^{-6} to 10^{-3} m). A fixed step size can also be set directly by the user.

- `rtol` / `atol`: Relative tolerance (default: `1e-9`) and absolute tolerance (default: `1e-25`) of the core differential equation solver.
- `sintering_energy_release_activated`: Set to `True` to account for the effect of coalescence energy release on nanoparticle temperature.
- `ambient_temp`: Ambient (room) temperature in K. Default: `293` K.
- `temperature_history`: Reactor temperature profile in K. For non-isothermal conditions, the temperature at different reactor positions must be defined in `simulation_setup.txt` as shown in Listing 4.3.

4.3.2 `MappingWrapper` and Classification Module Parameters

- `map_param`: Specifies the two process parameters to investigate in the 2D parametric map. Available options are `"P&T"`, `"T&l"`, `"l&d"`, and `"P&d"`, where P is pressure, T is temperature, l is reactor length, and d is the injected catalyst particle size.
- `grid_total`: Total number of grid points (i.e., $\sqrt{N} \times \sqrt{N}$) in the 2D map.
- `P_range_min` / `P_range_max`: Minimum and maximum pressure investigated in Pa.
- `P_iso`: Isobaric pressure in Pa, used when pressure is held constant in the map.
- `T_range_min` / `T_range_max`: Minimum and maximum temperature investigated in K.
- `T_iso`: Isothermal temperature in K, used when temperature is held constant in the map.
- `xdp_range_min` / `xdp_range_max`: Minimum and maximum primary particle size of the injected catalyst investigated in m.
- `xdp_const`: Fixed primary particle size in m, used when particle size is held constant in the map.

- `L_reactor_range_min` / `L_reactor_range_max`: Minimum and maximum total reactor length investigated in m.
- `L_reactor`: Fixed total reactor length in m, used when reactor length is held constant in the map.
- `xN_range_min` / `xN_range_max`: Minimum and maximum injected number concentration of nanoparticles in $\#/m^3$.
- `xN_const`: Fixed injected nanoparticle number concentration in $\#/m^3$.
- `ml_lambda_`: Regularization parameter controlling model complexity. Default: `0.001`.
- `ml_iterations`: Total number of training steps. Default: `10000`.
- `ml_alpha`: Regularization strength parameter. Default: `1.5`.
- `ml_post_cond`: Set to `True` (default) to activate the machine learning classification module, which post-processes the generated 2D map to identify SWCNT and MWCNT formation regions.

Chapter 5

Outputs

5.1 Description of output parameters

Using the instance obtained from either `GasReactor` or `MappingWrapper` class, the user has access to all simulation data for post-processing purposes. The available output parameters are described below:

- `AA12`: A dictionary containing the current gas composition, viscosity, density, specific heat, and molar fractions of the top three dominant hydrocarbon species.
- `D_cnt_saved`: A 2D array containing the mobility diameter of CNTs across bins at each length (time) step.
- `xdf_saved`: A 2D array containing the fractal dimension of agglomerates across bins at each length (time) step.
- `carbon_data`: A 1D array containing the carbon atom information for each bin at step i. The structure of this vector is as follows:

```
[# carbon_cnt,                # per agglomerate in #/kg_gas
 length of cnt,               # per agglomerate in m/kg_gas
 # carbon of graphene,        # per agglomerate in #/kg_gas
 area of graphene layer,      # total in m2/kg_gas
 # carbon inside,             # per agglomerate in #/kg_gas
 CNT condition]               # per bin: 0 = unsaturated, 1 =
    saturated
```

- `carbon_data_saved`: A 2D array containing `carbon_data` at each length (time) step.
- `carbon_data_saved_list`: A 1D array representing the mass balance check at each step. A value of `1` indicates that the total carbon mass balance in the reactor is preserved.
- `dp_saved`: A list of dictionaries containing the primary particle diameter at each length step.
- `temp_NP_saved`: An array containing the nanoparticle temperature at each length (time) step.
- `temp`: The current gas temperature in K.
- `wall_number_section_saved`: A 2D array containing the wall number of CNTs across bins at each length (time) step.
- `xdm_saved`: A 2D array containing the mobility diameter of agglomerates across bins at each length (time) step.
- `xk1`: A 1D array containing the Knudsen number across bins at each length (time) step.
- `xnpr_saved`: A 2D array containing the number of primary particles per agglomerate across bins at each length (time) step.
- `y`: A 1D array containing nanoparticle information for each bin at step i. The structure of this vector is as follows:

```
[N (vector), # number concentration of agglomerates per
   section
 V (vector), # volume concentration per agglomerate in m3/
    kg_gas
 A (vector), # total area concentration of all agglomerates
    per section in m2/kg_gas
 Carbon (vector), # total carbon per section
 C (scalar)]   # precursor gas concentration (for cases with
    no premade nanoparticles injected into the reactor)
```

5.2 Post-Processing

Since the user has full access to the `model` instance, any custom post-processing step is possible. However, `CarbonX` also provides a built-in module for post-processing as described below.

5.2.1 Mass Balance of Carbon and Metallic Nanoparticles

After solving the simulation, the built-in `Results_Processor` module can be used to evaluate the mass balance of carbon and metallic nanoparticles, as shown below:

Listing 5.1: Post-processing using the built-in `Results_Processor` module.

```
import Results_Processor

times, solutions = model.solve()

AA = Results_Processor.ResultsPostProcessor(model)

# Plot the psi-eta diagram with experimental data
AA.plot_psi_eta_diagram([0.004, 0.4, 1], add_experimental=True,
    regime_type='CR')

# Plot the carbon mass balance
fig, axes, data = AA.mass_balance_check()

# Plot the relative error
fig, ax, data = AA.plot_relative_error()
```

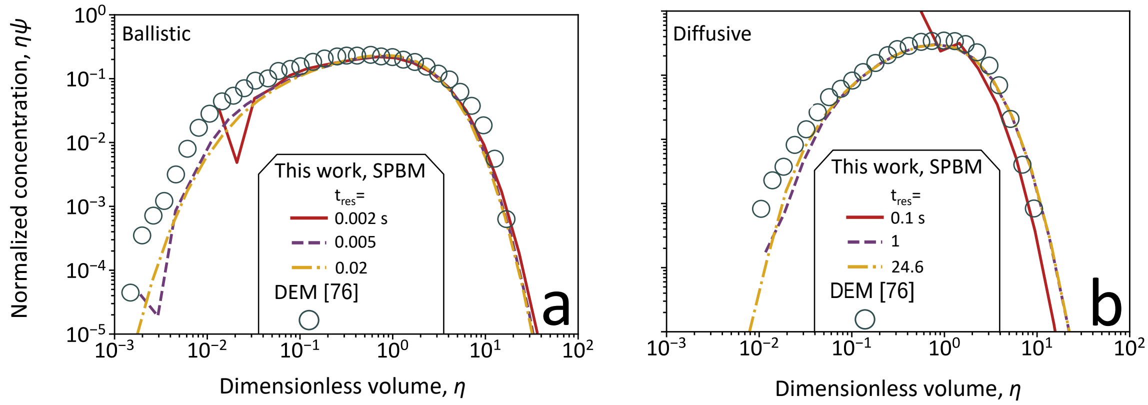

Figure 5.1: Self-preserving size distributions of coagulating nanoparticles with $\varphi_s = 10^{-4}$ in the ballistic (a: $d_{po} = 1$ nm) and diffusive (b: $d_{po} = 1000$ nm) regimes at 300 K and 1 atm.

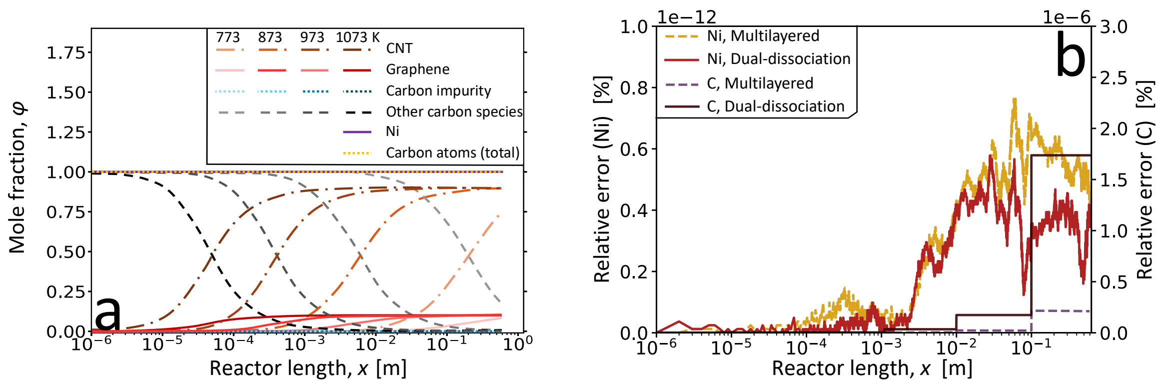

Figure 5.2: (a) The mole fractions of Ni and C in a flow reactor are shown as a function of reactor length at $f_{cnt} = 0.9$, corresponding to the condition where 90% of the carbon atoms precipitating at the CNT growth site (N_t) are converted into CNTs, while the remaining carbon atoms form graphene layers. The fractions of carbon present in CNTs (dash-dotted), graphene layers (solid), NP impurities (dotted), and NP-surface platelets (dashed) are compared at $T_{max} = 773$, 873, 973, and 1073 K. (b) The relative errors for a case study at 873 K, assuming multilayered and dual-dissociation models for Ni and C.

The `CarbonX` package has been benchmarked against available experimental measurements. Case studies and validation examples can be found in the package repository at https://github.com/Hsnrahbar/CarbonX_Package.

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